Final 1-Page Interview Revision Sheet

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DSA Patterns (Essentials)

- Sliding Window: grow right, shrink left; $O(n)$.
- Prefix Sum + HashMap: subarray sum problems.
- Two Pointers: sorted arrays, merging, partitioning.
- Binary Search: sorted or rotated arrays; $O(\log n)$.
- Heap: top-k problems; implemented as binary heap.
- DFS/BFS: trees and graphs; recursion vs queue.
- DP: overlapping subproblems; memo vs tabulation.

Time Complexity Quick Reference

- HashMap average: $O(1)$
- Sorting: $O(n \log n)$
- BFS/DFS: $O(V + E)$
- Binary Search: $O(\log n)$

Java Essentials

- **HashMap Internal:** buckets + hash + collision as list/tree.
- **ArrayList vs LinkedList:** random access vs insert/delete.
- **String immutable:** security, caching, thread-safety.
- **Streams:** map, filter, reduce; lazy evaluation.
- **volatile:** visibility guarantee, not atomicity.

Spring Boot Quick Notes

- DI/IoC: Spring manages object creation.
- @Component, @Service, @Repository: stereotype annotations.
- Filters vs Interceptors: request lifecycle vs handler lifecycle.
- RestController: returns JSON by default.
- Exception Handling: @ControllerAdvice.

System Design Mini Notes

- Cache (Redis): reduces DB load; TTL + eviction policy.
- Rate Limiter: token bucket in Redis; atomic counters.
- Message Queue: improves reliability; async processing.
- Metrics: p95 latency, throughput, error rate.
- Typical stack: Load Balancer → Service → Cache → DB.

Database Essentials

- Index: speeds reads, slows writes; B+ tree.
- Joins: inner, left, right.
- Transactions: ACID guarantees.
- MongoDB: document model, flexible schema.

Observability (New Relic Focus)

- Metrics: numerical time-series data.
- Logs: textual event records.
- Traces: follow request path across services.
- p95: 95% of requests are faster than this value.